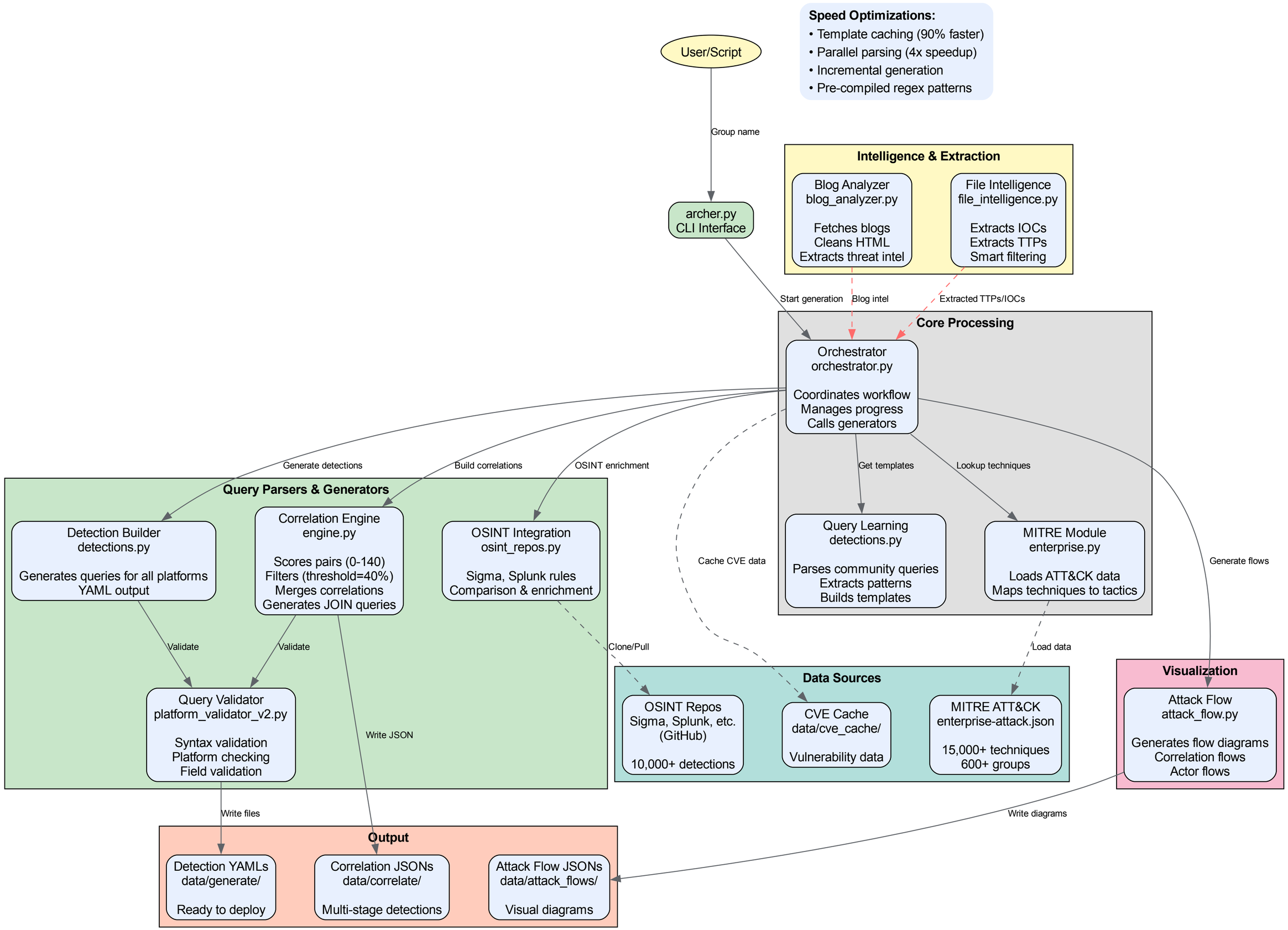

Speed Optimizations:
• Template caching (90% faster)
• Parallel parsing (4x speedup)
• Incremental generation
• Pre-compiled regex patterns
User/Script
Group name
archer.py
CLI Interface
Intelligence & Extraction
Blog Analyzer
blog_analyzer.py
Fetches blogs
Cleans HTML
Extracts threat intel
File Intelligence
file_intelligence.py
Extracts IOCs
Extracts TTPs
Smart filtering
Start generation
Blog intel
Extracted TTPs/IOCs
Core Processing
Orchestrator
orchestrator.py
Coordinates workflow
Manages progress
Calls generators
Generate detections
Build correlations
OSINT enrichment
Get templates
Lookup techniques
Query Parsers & Generators
Detection Builder
detections.py
Generates queries for all platforms
YAML output
Correlation Engine
engine.py
Scores pairs (0-140)
Filters (threshold=40%)
Merges correlations
Generates JOIN queries
OSINT Integration
osint_repos.py
Sigma, Splunk rules
Comparison & enrichment
Query Learning
detections.py
Parses community queries
Extracts patterns
Builds templates
MITRE Module
enterprise.py
Loads ATT&CK data
Maps techniques to tactics
Cache CVE data
Generate flows
Validate
Validate
Clone/Pull
Load data
Query Validator
platform_validator_v2.py
Syntax validation
Platform checking
Field validation
Write JSON
Data Sources
OSINT Repos
Sigma, Splunk, etc.
(GitHub)
10,000+ detections
CVE Cache
data/cve_cache/
Vulnerability data
MITRE ATT&CK
enterprise-attack.json
15,000+ techniques
600+ groups
Visualization
Attack Flow
attack_flow.py
Generates flow diagrams
Correlation flows
Actor flows
Write files
Write diagrams
Output
Detection YAMLs
data/generate/
Ready to deploy
Correlation JSONs
data/correlate/
Multi-stage detections
Attack Flow JSONs
data/attack_flows/
Visual diagrams